ASP.NET Core Environment Setup

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In this article, I will discuss the ASP.NET Core Environment Setup required for developing ASP.NET Core Applications. Please read our previous article before proceeding, where we briefly introduced the ASP.NET Core Framework. Let's start the ASP.NET Core Environment Setup required for developing ASP.NET Core Applications on Windows OS. Let's understand what necessary and optional components required for ASP.NET Core Application Development.

Operating System:

ASP.NET Core development can be done on any modern operating system, such as Windows, macOS, or Linux. Let's use the Windows Operating System in the Course.

- **Windows:** Microsoft Visual C# .NET development, as it fully supports Visual Studio and .NET 4.5 also allows easy integration with Microsoft Entity Server.
- **Linux:** Linux is the most popular platform for .NET development, but keep in mind that some Microsoft-specific tools like SQL Server Management Studio are not available natively on Linux platforms.

Integrated Development Environment (IDE):

You can choose from various IDEs for ASP.NET Core development including:

- **Visual Studio:** Highly recommended for Windows users. It provides comprehensive support for .NET development, including project templates, a powerful debugger, and direct integration with Entity Server Management Studio for database management and development.
- **Visual Studio Code:** Visual Studio Code is a lightweight, cross-platform code editor supporting ASP.NET Core development. It offers excellent extensions support, integrated terminal, and source control integration. Visual Studio Code is available for Windows, macOS, and Linux.

.NET SDK:

This is the software development kit for developing and running applications with ASP.NET Core. It includes the .NET runtime, libraries, and command-line tools.

ASP.NET Core:

ASP.NET Core is the framework for building web applications, including Web APIs, with .NET Core. It is included in the .NET SDK. The latest stable version of ASP.NET Core at the time of this tutorial is Core 2.2. Keep in mind that, Either we install the .NET SDK or ASP.NET Core, the latter will include the necessary libraries for Web development.

Database Management System:

You need a database management system depending on your application's requirements. Common options for ASP.NET Core include:

- **Microsoft SQL Server:** This is a popular choice for Windows environments, especially in enterprise applications. It offers robust performance and integration with Microsoft tools.
- **MySQL or PostgreSQL:** Cross-platform, open-source relational database management systems. These are suitable for Linux and macOS environments.
- **SQLite:** It is a lightweight and file-based database system suitable for development and small-scale applications.

SQL Server Management Studio (SSMS):

SQL Server Management Studio (SSMS) is an integrated environment for managing SQL Server databases. It offers tools for configuration, administration, and design.

.NET Client:

- **PostmanPostman:** This is a versatile API client that works on Windows, macOS, and Linux. It's helpful for testing and troubleshooting APIs developed with ASP.NET Core.
- **Swagger (OpenAPI):** Integrating Swagger into your ASP.NET Core project allows you to generate interactive API documentation and automatically test API calls directly from your web browser.

These are the essential software components required for ASP.NET Core.NET and other IDE development. In the next article, I will discuss the Installation of Visual Studio Code, Visual Studio, and ASP.NET Core Framework. I hope you like this article. Please leave feedback for **ASP.NET Core Environment Setup**. Hope you enjoy the ASP.NET Core Environment Setup article.

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